

Member 1

- **Name:** Ashutosh Mohapatra
- **Roll number:** 211020003

Member 2

- **Name:** Taran Poojari
- **Roll number:** 211020004

Member 3

- **Name:** Mohammed Wafeeq Kazi
- **Roll number:** 211020022

Assignment 1

Design of Machine Elements Project

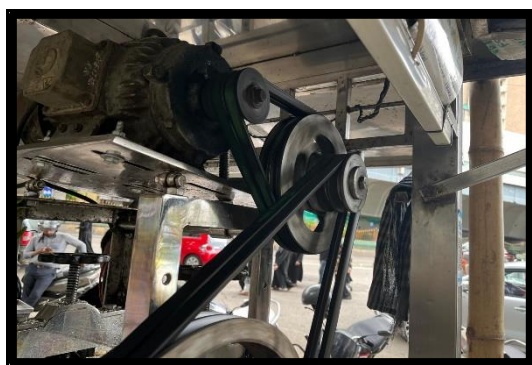
Design and Analysis of a Conventional Sugarcane Juicer Machine

The report outlines the following points –

- 1. Introduction.*
- 2. Initial layout as per motor specifications.*
- 3. Mounting structure, its Analysis*
- 4. layout changes.*
- 5. MATLAB Program for gear calculations.*
- 6. Structural Analysis.*
- 7. Bill Of Quantities.*
- 8. Amount of sugarcane juice collected in a day.*
- 9. Conclusion.*

Introduction:

- *Sugarcane juicer machines are one of the most common machines manufactured industrially due to the demands and high consumption of the conventional beverage in urban and suburban regions.*
- *The layout of a basic machine consists of a pulley system driven by an electric motor, which is further connected to a gear system, to which the final rollers are connected to obtain significant speed reduction and crushing power.*
- *The distance between the rollers is 10-15 mm, which is smaller than the diameter of the sugarcane itself, else the sufficient crushing force would not be developed while extracting the juice from the stalk.*
- *The roller distance can be varied with the help of a simple mechanism operated by a lead screw-based mechanism. For simplicity, the mechanism is not added in the project and only the fundamental elements are described in the given report. The pictures of a standard sugar cane juicer machine are below:*



Initial Layout

- The motor specifications used for this project are as follows:

Type:	3 - Phase Electric motor:
Power:	1.4 HP
Speed:	630 rpm

- Initial Model:

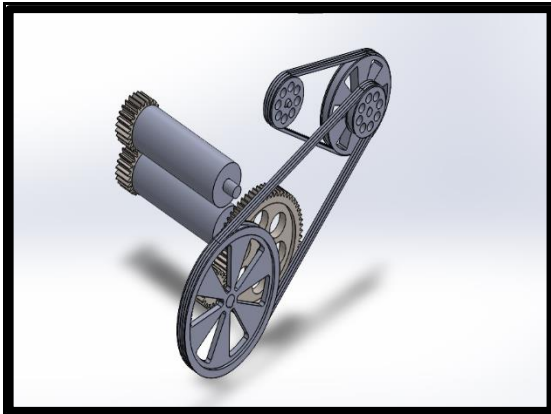


Fig: Isometric View.

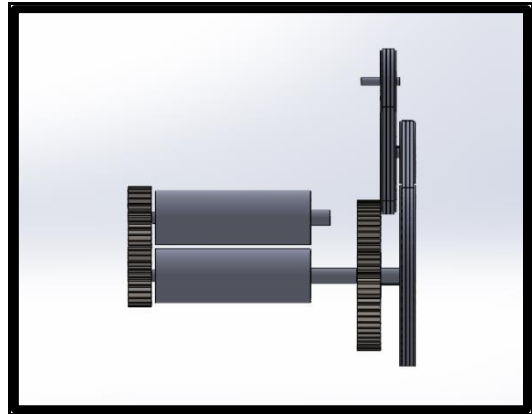


Fig: Front View.

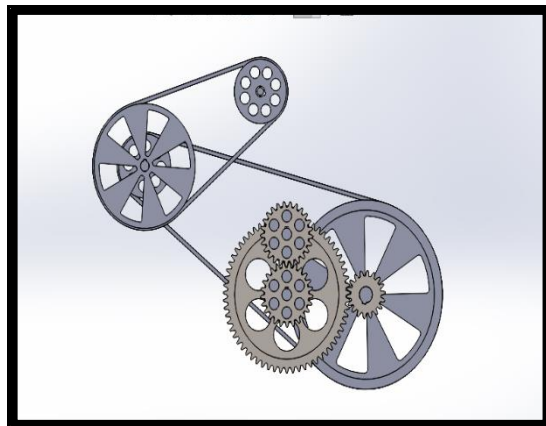


Fig: Left View

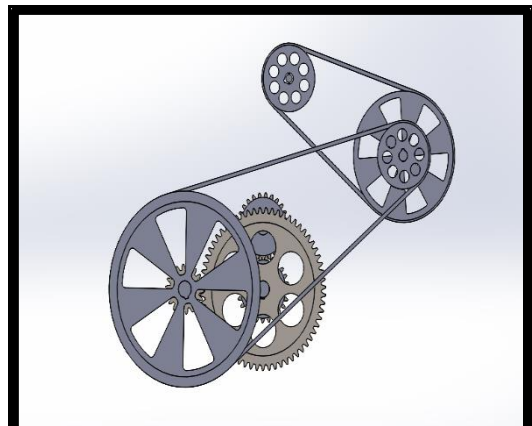


Fig: Right View

- *Based on the calculations, the initial layout design was created in SolidWorks as follows.*
- *The supporting calculations are attached at the end of the report for both the gear and pulley systems.*

SUPPORT STRUCTURE



- *The sugarcane machine features a support structure designed to hold the pulleys, gears, motors, and bearings securely in place. The structure includes four designated areas where rotating components are installed, each supported by bearings to ensure smooth operation.*
- *The upper base of the structure is responsible for supporting the motor, while the lower base incorporates a collection mechanism. When the sugarcane is crushed, the extracted juice flows into a container below. A strainer in this mechanism prevents sugarcane fibre's from mixing with the juice.*
- *In this setup, the primary load is borne by the side walls of the structure. The loads on the collection chamber and motor base are considered minimal and not significant enough to impact the overall structural integrity.*
- *Therefore, we will be doing the Analysis of the side walls to ensure the stability of the structure.*

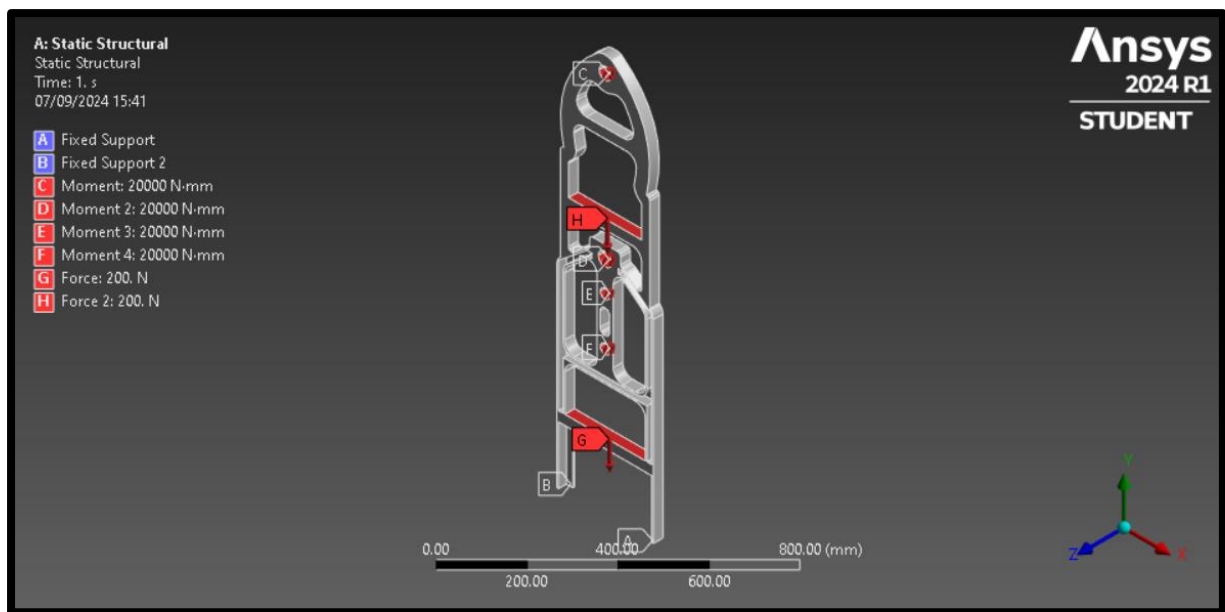
Analysis of static structure which provides support to the crushing mechanism of the sugarcane machine:

➤ **Material Used:**

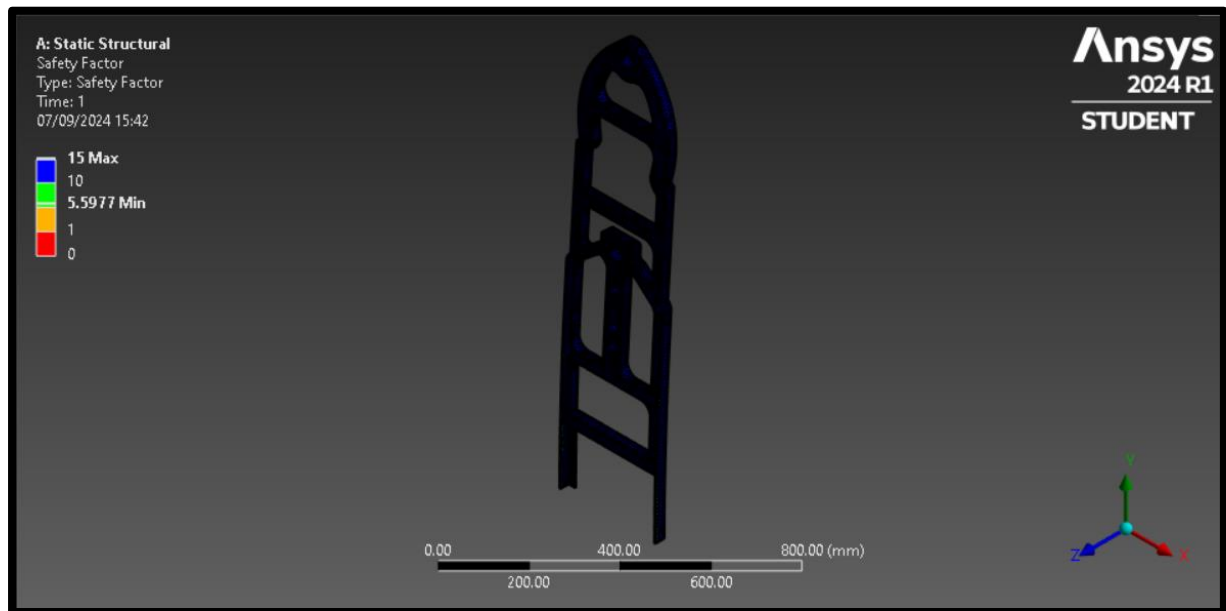
- *Structural Steel is used for this purpose for providing robust support to the mechanisms.*
- $E = 200 \text{ GPa}$.
- $P.R = 0.3$.

➤ **Analysis Settings:**

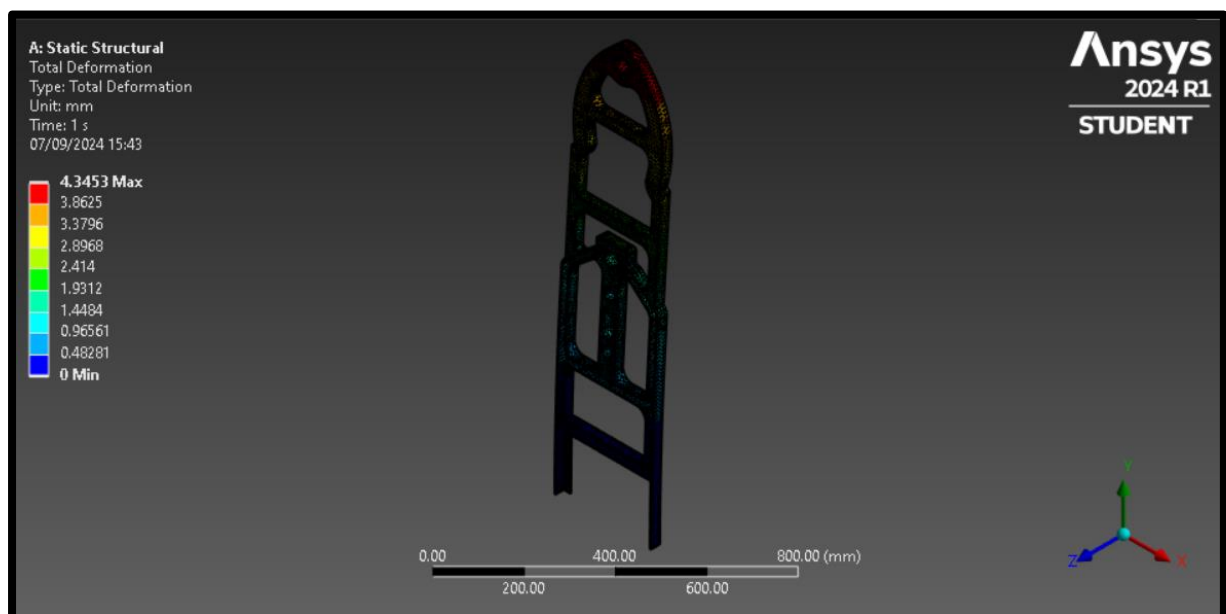
- **A:** Fixed to the ground.
- **B:** Fixed to the ground.
- **C:** Moment due to presence of gear/ pulley.
- **D:** Moment due to presence of gear/ pulley.
- **E:** Moment due to presence of gear/ pulley.
- **F:** Moment due to presence of gear/ pulley.
- **G:** Load due to Collection mechanism.
- **H:** load due to motor.
- **Note:** Exaggerated loads and moments are applied as a factor of safety.



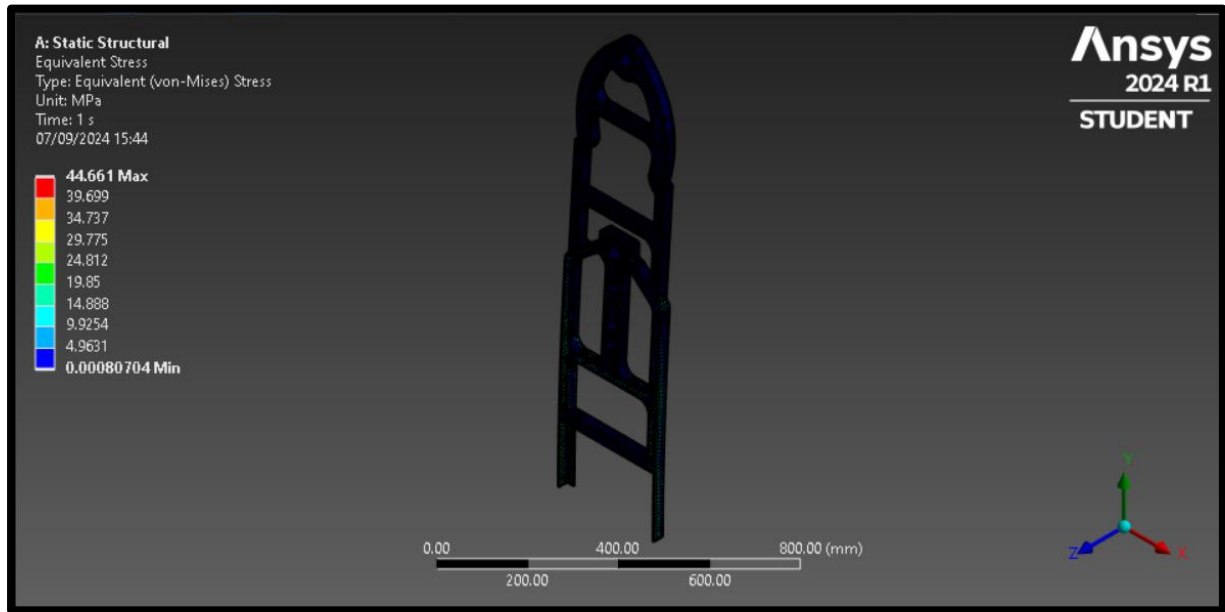
➤ **Factor of Safety:**



➤ **Deformation:**



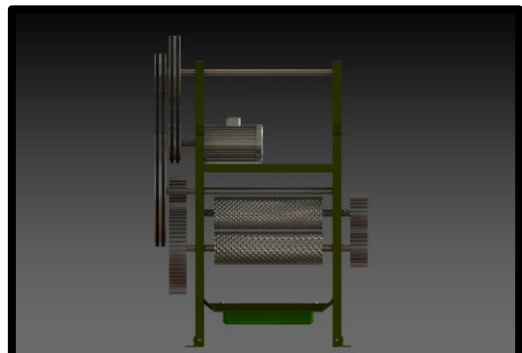
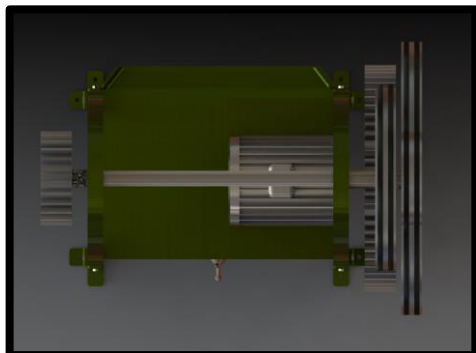
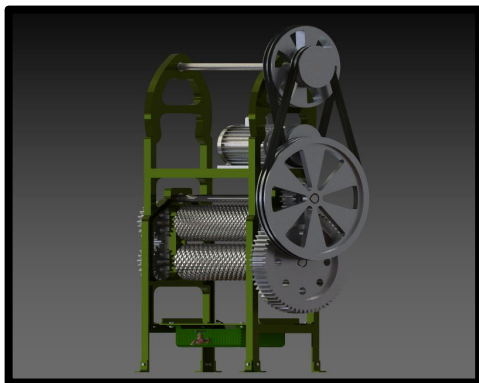
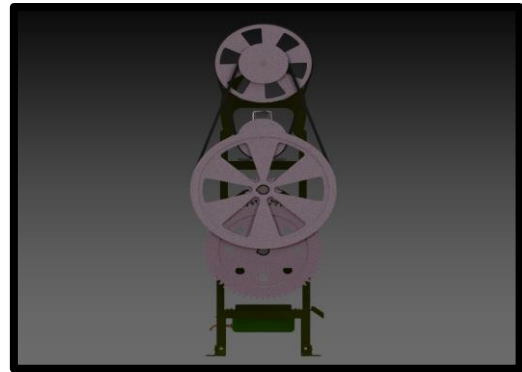
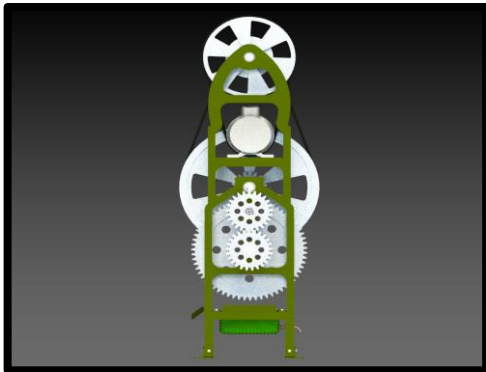
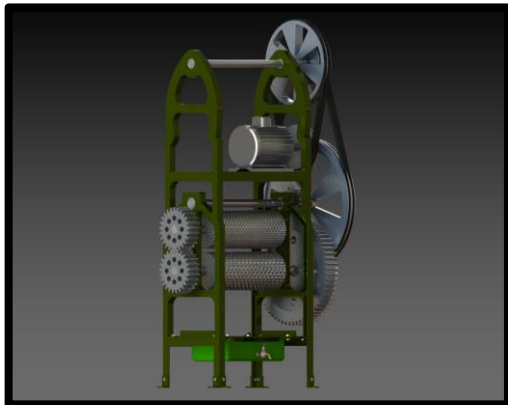
➤ **Equivalent Stress:**



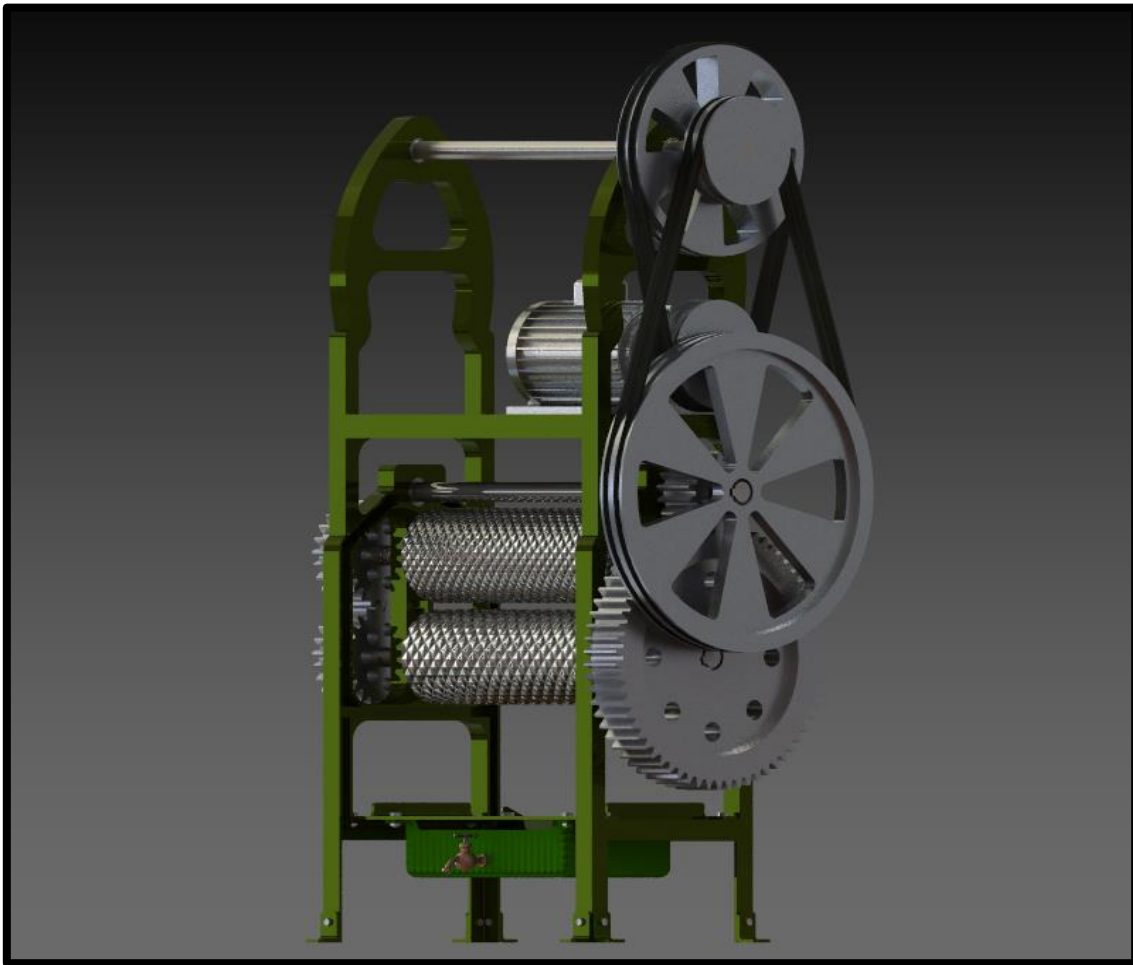
➤ **Results:**

- *The mechanism will be able to sustain the applied load as the factor of safety is 5.57.*
 - *The gears are supported using two side walls and the load is unevenly distributed on both sides.*
 - *The analysis done is of the wall experiencing greater load. So therefore, we can say that the side experiencing less load in comparison to other side will also sustain.*
-

LAYOUT CHANGES / FINAL CAD



Final CAD



- The Mechanical System Comprising of pulleys, gears, rollers, support structure and motor where connected/assembled in such a way so as to provide a compact structure.
- The dimensions of the Sugarcane machine are roughly
 - **Height:** 1.2 meter.
 - **Breadth:** 0.35 meter.
 - **Length:** 0.5 meter.
- The Components which will rotate are provided with bearings to provide a smooth frictionless Rotations.
- The Gears and Pulleys are mated with suitable keys to provide stable locking with interconnected shafts.
- The Rollers are provided with grooves which amplifies the crushing effect.
- A tap is fitted on the container, providing feasible collection of crushed juice.

MATLAB PROGRAM

- A program was written in MATLAB for convenience in the gear calculator in case of any changes in parameters (dimensional or material-based properties), and thus, repeated hand calculations were avoided. The results of the program for the given gear systems are given below –

```
Spur_Gear.m  Bevel_Gear.m  +
1  p = input("Enter power transmitted in kW: ");
2  N = input("Enter input speed in rpm: ");
3  i = input("Enter gear ratio: ");
4  p_sig_y = input("Enter yield strength of pinion material in N/mm^2: ");
5  g_sig_y = input("Enter yield strength of gear material N/mm^2: ");
6  p_sig_u = input("Enter ultimate tensile strength of pinion material N/mm^2: ");
7  g_sig_u = input("Enter ultimate tensile strength of gear material N/mm^2: ");
8  p_bhn = input("Enter brinell hardness of pinion material: ");
9  g_bhn = input("Enter brinell hardness of gear material: ");
10 % For alloy steels, with no heat treatment
11 kb = 1;
12 k_sig = 1.5;
13 n = 2.5;
14 z1 = 18;
15 z2 = i*z1;
16 z2 = z2+1;
17 sig_1_p = 0.35*p_sig_u*10+1200;
18 sig_1_g = 0.35*g_sig_u*10+1200;
19 sig_b_p = (1.4*kb*sig_1_p)/(n*k_sig);
20 sig_b_g = (1.4*kb*sig_1_g)/(n*k_sig);
21 y_p = (0.154-(0.912/z1))*pi;
22 y_g = (0.154-(0.912/z2))*pi;
23 lp = sig_b_p*y_p;
24 lg = sig_b_g*y_g;
25 if(lp<lg)
26     disp("Pinion is weaker than gear and is to be designed for bending and wear strength.");
27     mt = 1.3*((p*1000*60)/(2*pi*N))*10;
```

```
Spur_Gear.m  Bevel_Gear.m  +
26     disp("Pinion is weaker than gear and is to be designed for bending and wear strength.");
27     mt = 1.3*((p*1000*60)/(2*pi*N))*10;
28     psi = 10;
29     m = 1.26*(2.5*mt/(y_p*sig_b_p*psi*z1))^(1/3); % Designing for higher torque
30     mn = 1.3*m*10;
31     disp("Module in mm: ");
32     mm = round(mn);
33     disp(mm);
34     disp("Face Width in mm: ");
35     b = psi*mm;
36     disp(b);
37     disp("Center distance in mm: ");
38     a = mm*(z1+z2)*0.5;
39     disp(a);
40     E = 2.15*10^(6);
41     disp("Actual gear ratio: ");
42     i_act = z2/z1;
43     disp(i_act);
44     disp("Induced compressive stress in kgf/cm^2: ");
45     sig_c_ind = 0.74*((i_act+1)/(a*0.1))*((i_act+1)/(i_act*b*0.1)*(E*mt))^(1/2);
46     disp(sig_c_ind);
47     cb = 25;
48     disp("Compressive strength in kgf/cm^2: ");
49     sig_c = kb*cb*p_bhn;
50     disp(sig_c);
51     if(sig_c<sig_c_ind)
```

```

Spur_Gear.m  Bevel_Gear.m  +
50     disp(sig_c);
51     if(sig_c<sig_c_ind)
52         disp("Pinion is not safe in wear, change hardness to following value: ");
53         bp = (sig_c_ind)/(kb*cb);
54         disp(bp);
55     end
56
57     if(sig_c>sig_c_ind)
58         disp("Pinion is safe in wear.");
59     end
60
61     disp("Induced bending stress in kgf/cm^2: ");
62     sig_b_ind = (i_act+1)/(a*0.1*mm*0.1*b*0.1*y_p)*mt;
63     disp(sig_b_ind);
64     disp("Pinion bending strength in kgf/cm^2: ");
65     disp(sig_b_p);
66
67     if(sig_b_ind<sig_b_p)
68         disp("Pinion is safe in bending.");
69     end
70
71     if(sig_b_ind>sig_b_p)
72         disp("Pinion is unsafe in bending. Increase module to higher value.");
73     end
74 end
75
76 if(lp>lg)

```

```

Spur_Gear.m  Bevel_Gear.m  +
76     if(lp>lg)
77         disp("Gear is weaker than pinion and is to be designed for bending and wear strength.");
78         mt = 1.3*((p*1000*60)/(2*pi*(N)))^10;
79         psi = 10;
80         m = 1.26*(2.5*mt/(y_g*sig_b_g*psi*z2))^(1/3);
81         mn = 1.3*m*10;
82         disp("Module in mm: ");
83         mm = round(mn);
84         disp(mm);
85         disp("Face Width in mm: ");
86         b = psi*mm;
87         disp(b);
88         disp("Center distance in mm: ");
89         a = mm*(z1+z2)*0.5;
90         disp(a);
91         E = 2.15*10^6;
92         disp("Actual gear ratio: ");
93         i_act = z2/z1;
94         disp(i_act);
95         disp("Induced compressive stress in kgf/cm^2: ");
96         sig_c_ind = 0.74*((i_act+1)/(a*0.1))*((i_act+1)/(i_act*b*0.1)*(E*mt))^(1/2);
97         disp(sig_c_ind);
98         cb = 25;
99         disp("Compressive strength in kgf/cm^2: ");
100        sig_c = kb*cb*g_bhn;
101        disp(sig_c);

```

```

Spur_Gear.m  Bevel_Gear.m  +
99     disp("Compressive strength in kgf/cm^2: ");
100    sig_c = kb*cb*g_bhn;
101    disp(sig_c);
102    if(sig_c<sig_c_ind)
103        disp("Gear is not safe in wear, change hardness to following value: ");
104        bg = (sig_c_ind)/(kb*cb);
105        disp(bg);
106    end
107
108    if(sig_c>sig_c_ind)
109        disp("Gear is safe in wear.");
110    end
111
112    disp("Induced bending stress in kgf/cm^2: ");
113    sig_b_ind = (i_act+1)/(a*0.1*mm*0.1*b*0.1*y_g)*mt;
114    disp(sig_b_ind);
115    disp("Gear bending strength in kgf/cm^2: ");
116    disp(sig_b_g);
117
118    if(sig_b_ind<sig_b_g)
119        disp("Gear is safe in bending.");
120    end
121
122    if(sig_b_ind>sig_b_g)
123        disp("Gear is unsafe in bending. Increase module to higher value.");
124    end
125 end

```

➤ Results:

```
>> Spur_Gear
Enter power transmitted in kW: 1.0444
Enter input speed in rpm: 105
Enter gear ratio: 3.5
Enter yield strength of pinion material in N/mm^2: 380
Enter yield strength of gear material N/mm^2: 310
Enter ultimate tensile strength of pinion material N/mm^2: 720
Enter ultimate tensile strength of gear material N/mm^2: 570
Enter brinell hardness of pinion material: 241
Enter brinell hardness of gear material: 187
Pinion is weaker than gear and is to be designed for bending and wear strength.
Module in mm:
6

Face Width in mm:
60

Center distance in mm:
246

Actual gear ratio:
3.5556

Induced compressive stress in kgf/cm^2:
3.2628e+03
```

```
Compressive strength in kgf/cm^2:
6025

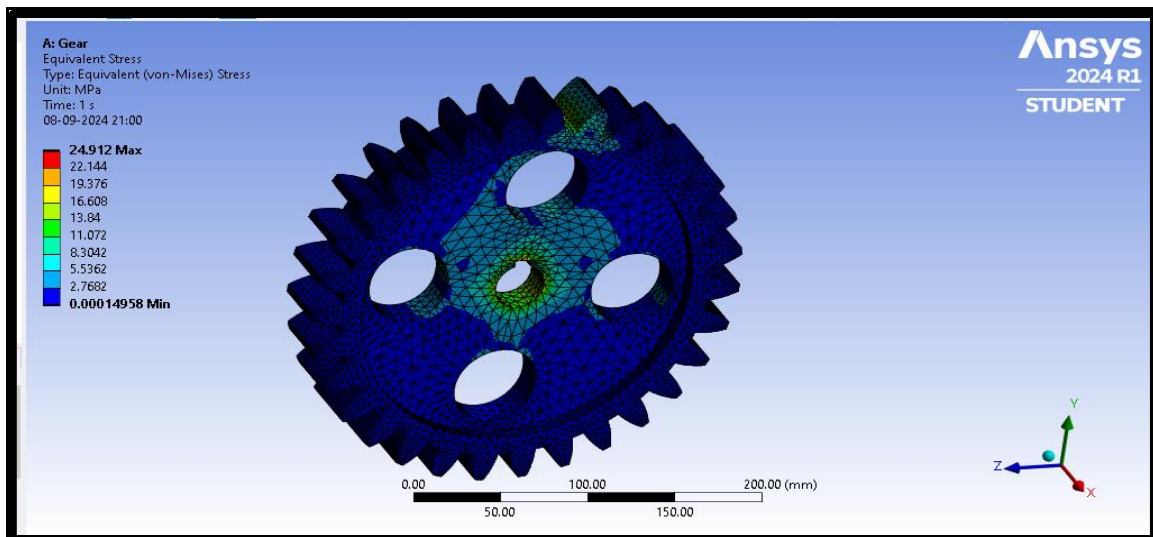
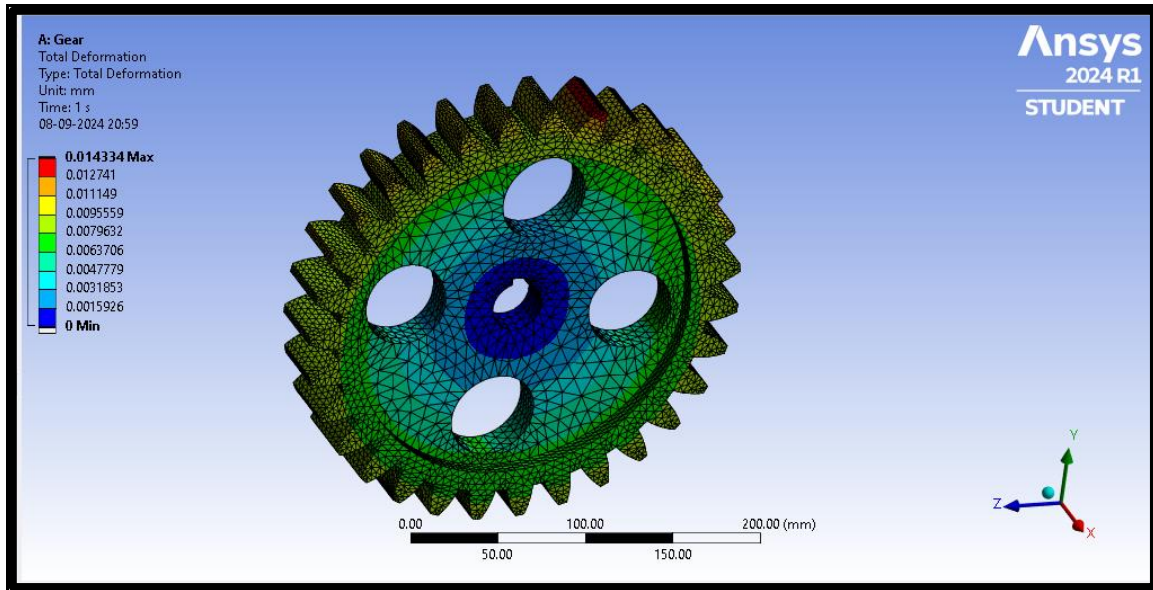
Pinion is safe in wear.
Induced bending stress in kgf/cm^2:
195.6617

Pinion bending strength in kgf/cm^2:
1.3888e+03

Pinion is safe in bending.
```

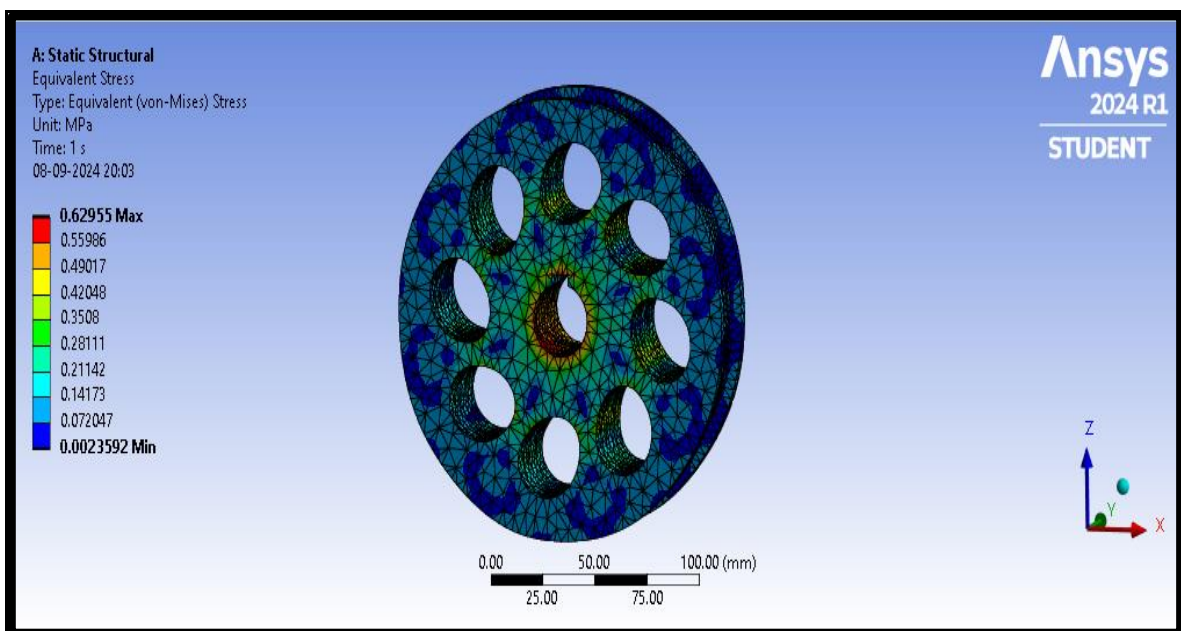
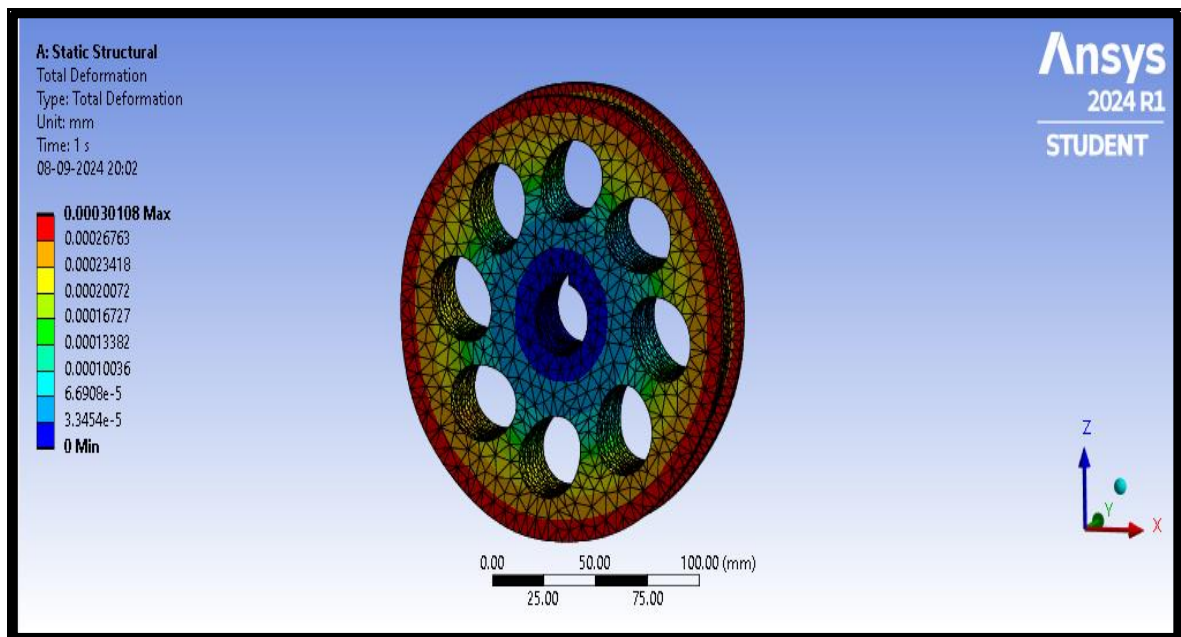
Structural Analysis of Gear Pulleys and Shaft.

➤ Placeholder (Support) gear



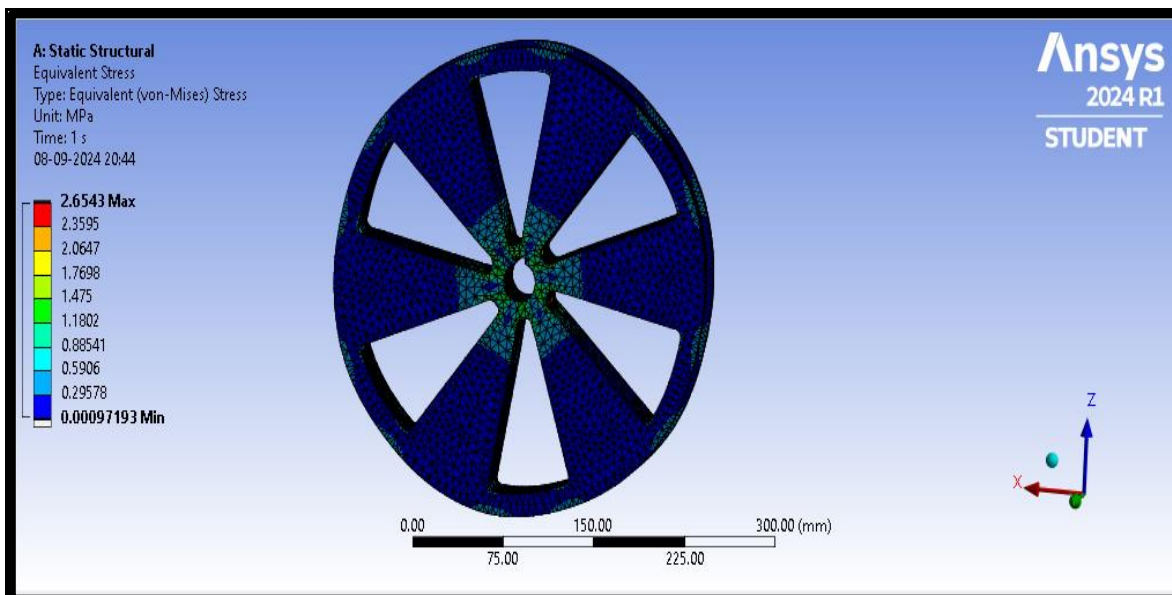
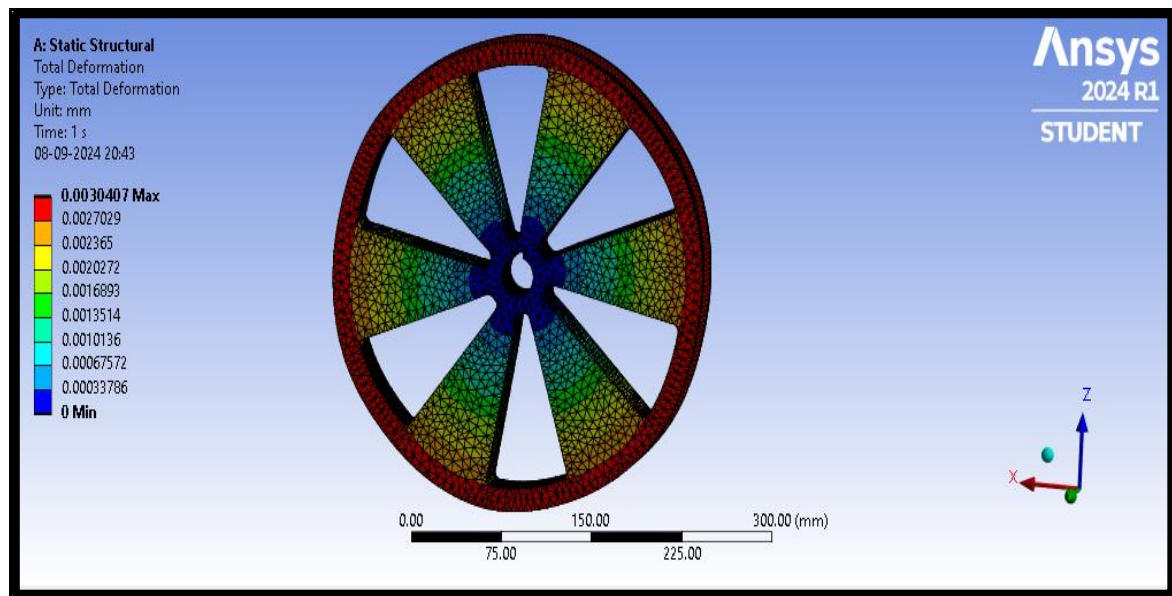
Material	C-50
Force (on tooth)	2500 N
Maximum Deformation	0.014334 mm
Maximum Induced Stress (Von Mises)	24.912 MPa
Factor of Safety	15

➤ Motor Pulley



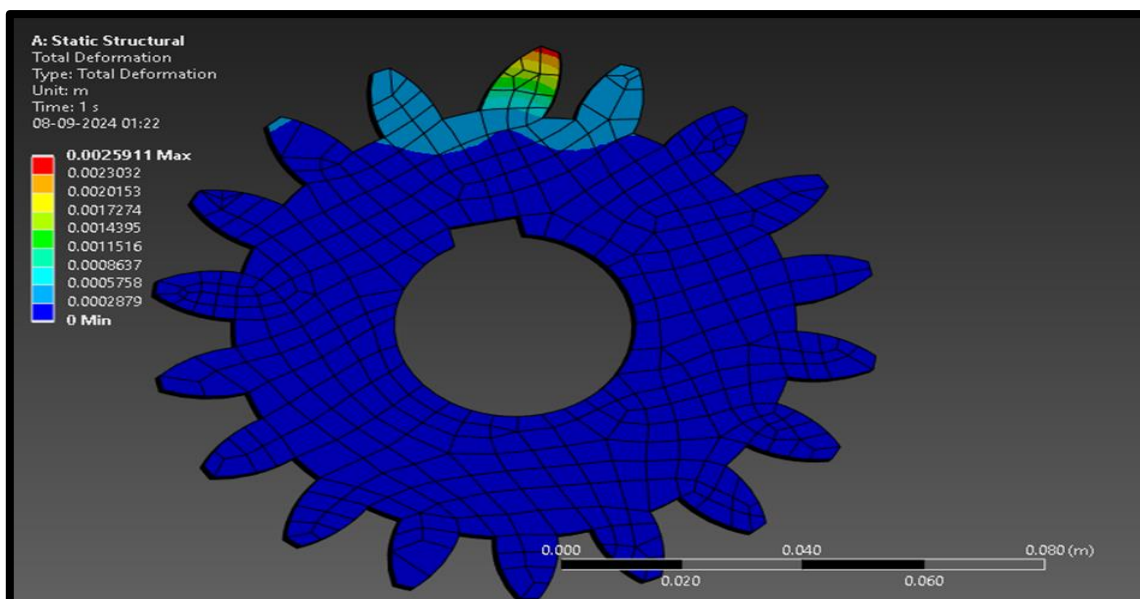
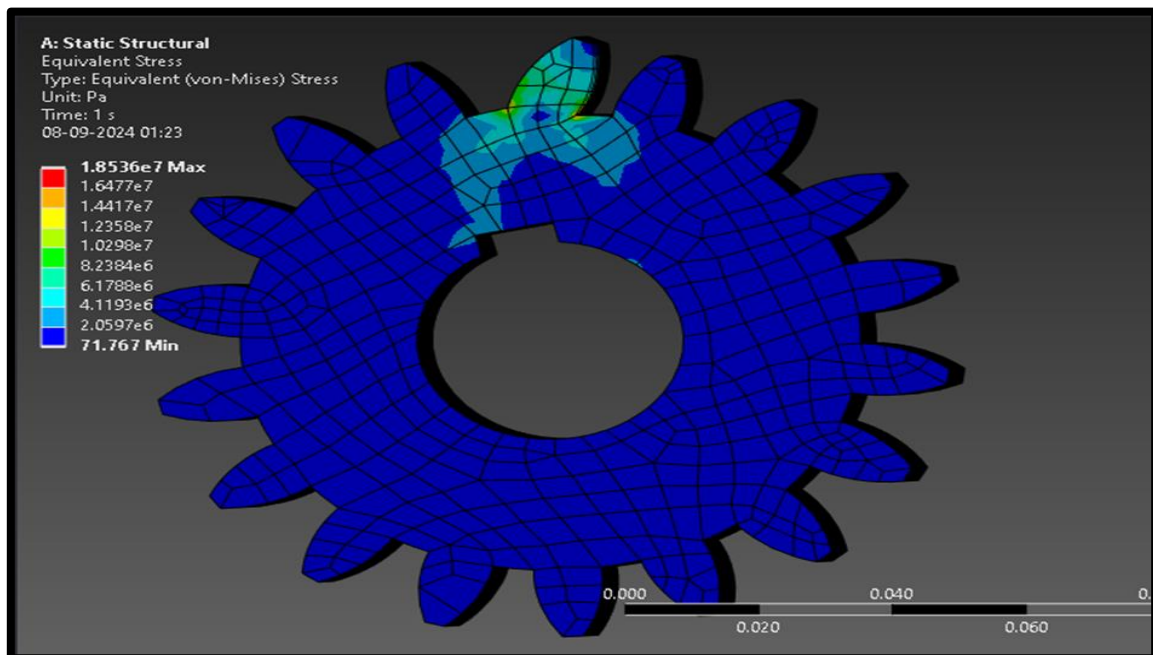
Material	Structural Steel
Moment	15831 N.mm
Maximum Deformation	0.00030108 mm
Maximum Induced Stress (Von Mises)	0.62955 MPa
Factor of Safety	15

➤ *Second Pulley*



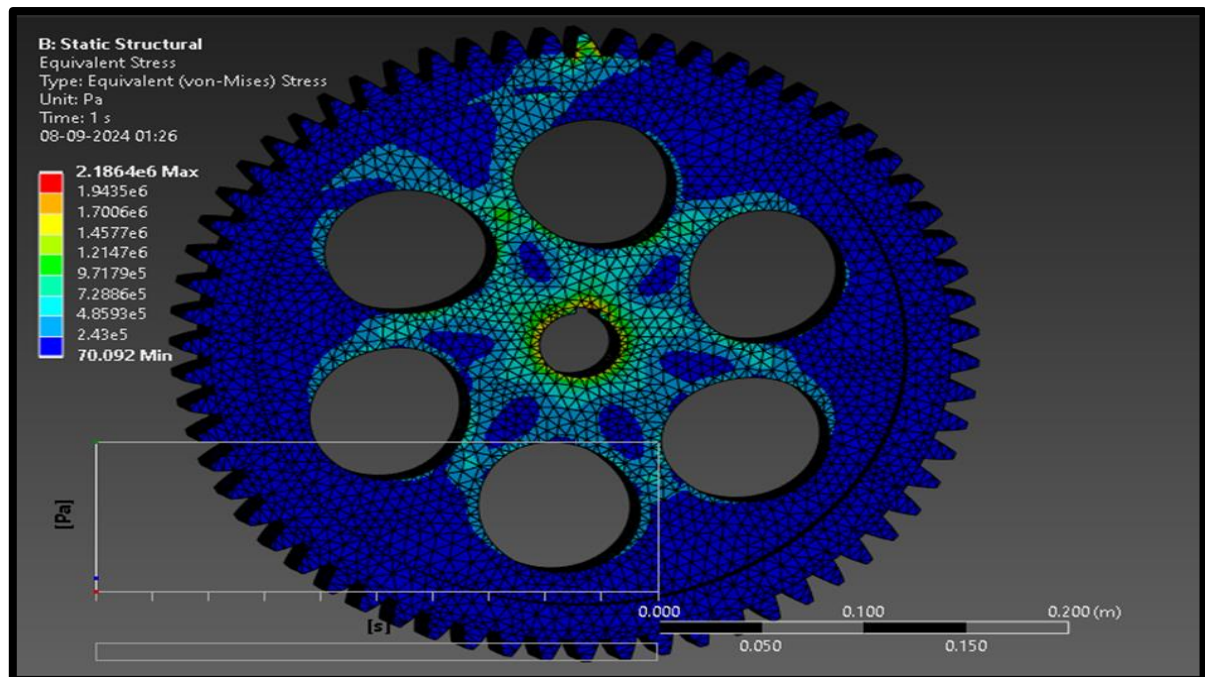
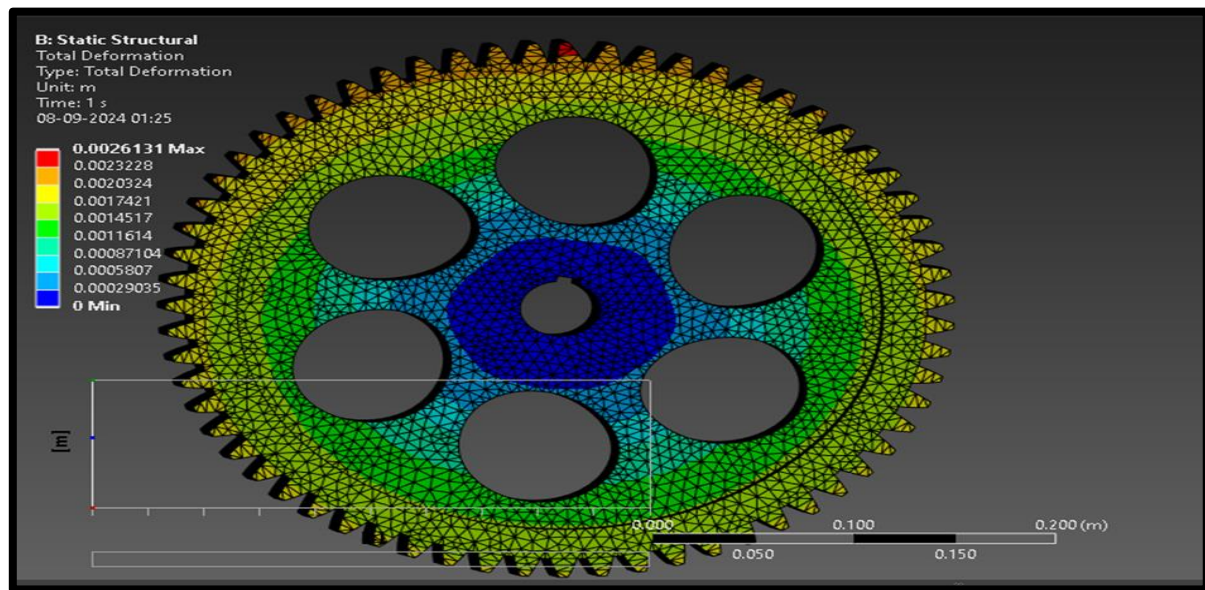
<i>Material</i>	<i>Structural Steel</i>
<i>Moment</i>	<i>31662 N.mm</i>
<i>Maximum Deformation</i>	<i>0.0030407 mm</i>
<i>Maximum Induced Stress (Von Mises)</i>	<i>2.6543 MPa</i>
<i>Factor of Safety</i>	<i>15</i>

➤ Pinion



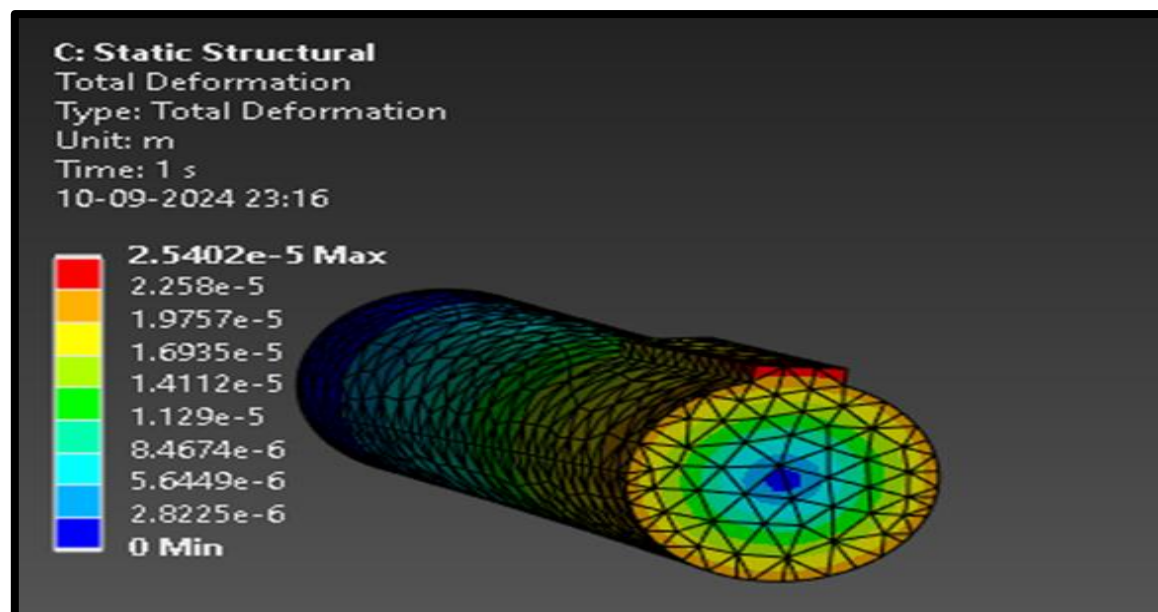
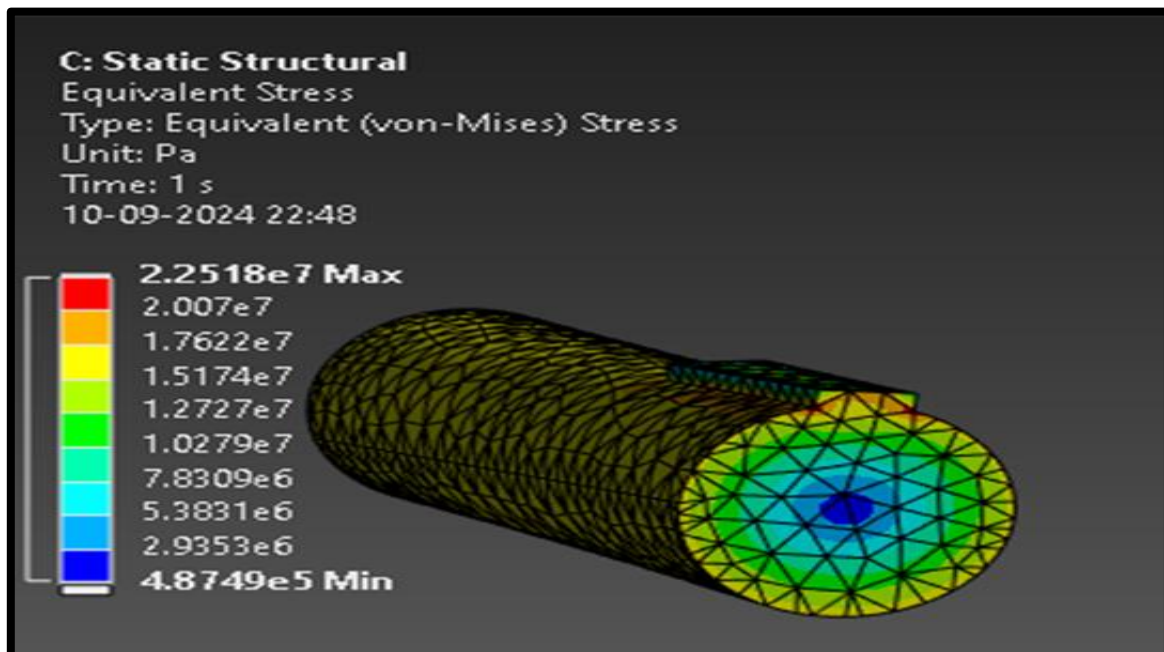
Material	C-50
Force (on tooth)	2288.963 N
Maximum Deformation	0.0026 m
Maximum Induced Stress (Von Mises)	18 MPa
Factor of Safety	15

➤ Gear



Material	C-35
Force (on tooth)	2288.963 N
Maximum Deformation	0.0026 m
Maximum Induced Stress (Von Mises)	2.15 MPa
Factor of Safety	15

➤ *Roller shaft*



<i>Material</i>	20MnCr5
<i>Moment</i>	123.604038 Nm
<i>Maximum Deformation</i>	25.4 m
<i>Maximum Induced Stress (Von Mises)</i>	22.52 MPa
<i>Factor of Safety</i>	15

Bill Of Materials

PART NAME	MATERIAL	QUANTITY	COST PER PIECE	TOTAL COST
<i>Pulleys (stage-1).</i>	<i>Structural Steel</i>	<i>1</i>	<i>₹4,000</i>	<i>₹4,000</i>
<i>Pulleys (stage-2).</i>	<i>Structural Steel</i>	<i>1</i>	<i>₹4,000</i>	<i>₹4,000</i>
<i>Pinion Gear.</i>	<i>C-50</i>	<i>1</i>	<i>₹5,000</i>	<i>₹5,000</i>
<i>Main Gear.</i>	<i>C-35</i>	<i>1</i>	<i>₹4,000</i>	<i>₹4,000</i>
<i>Placeholder Gear.</i>	<i>C-50</i>	<i>1</i>	<i>₹4,500</i>	<i>₹4,500</i>
<i>Shaft.</i>	<i>C-45</i>	<i>4</i>	<i>₹1500</i>	<i>₹6,000</i>
<i>Key.</i>	<i>C-45</i>	<i>8</i>	<i>₹250</i>	<i>₹2,000</i>
<i>Belt.</i>	<i>Canvas</i>	<i>4</i>	<i>₹625</i>	<i>₹2500</i>
<i>Nuts and Bolts.</i>		<i>30</i>	<i>₹10</i>	<i>₹300</i>
<i>Bearings.</i>		<i>16</i>	<i>₹500</i>	<i>₹8000</i>
<i>Motors.</i>		<i>1</i>	<i>₹6000</i>	<i>₹6000</i>

- **Total cost:** ₹46300.
- **Total cost with added miscellaneous costs:** ₹50,000 Approx.
- *It is important to note that these costs are Approximate costs and are can be subjected to change.*

Number of cups filled on an average:

- **Working hours:** 7 hours.
- **Worker efficiency:** 70 percent.
- **Capacity of machine:** 20 kg/hour.
- **Amount of juice extracted from one sugarcane:** 250ml/kg.
- **Effective operating time is:** $7 \times 0.7 = \underline{4.9 \text{ hours.}}$
- **Total sugarcane crushed in a day:** $4.9 \times 20 = \underline{98 \text{ kg.}}$
- **Total juice collected:** $98 \times 250 = \underline{24500 \text{ ml.}}$
- **Number of cups fill:** $24500 / 250 = \underline{98 \text{ cups.}}$
- Number of customers are not taken into consideration as amount of juice extracted depends on the number of customers visited.
- This calculation was done to find out how much on an average a sugarcane vendor can extract.

Conclusion.

- The design of the roadside sugarcane machine effectively balances portability, ease of operation, and juice extraction efficiency. With a capacity of 20 kg/hour and an average juice yield of 250 ml per kg, the machine is suitable for single-labourer operation, typical of street vending environments. Over the course of a 7-hour workday, with a labour efficiency of 70%, the machine processes approximately 98 kg of sugarcane, producing 24.5 liters of juice, which is sufficient to fill 98 cups of 250 ml each.
- The machine's performance is optimized for manual operation, taking into account the worker's multitasking duties and potential downtimes. The moderate capacity ensures that the machine is manageable for a single operator, allowing for efficient throughput without overwhelming the worker. Additionally, the machine's design is robust enough for continuous use, but compact and simple enough to require minimal maintenance in roadside environments.
- This design is ideal for small-scale sugarcane juice vendors, ensuring a balance between production efficiency and labour costs.